Proving triangles congruent

Core Concepts: Proving triangles congruent

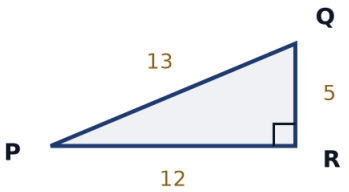
1 In $\triangle ABC$ studying proving triangles congruent, $m\angle A = 31^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 104°

b. 114°

c. 96°

d. 149°



2 A geometric figure scaled by factor $k = 2$ has an original side length of 5 cm. What is the corresponding image length?

a. 8 cm

b. 10 cm

c. 15 cm

d. 20 cm

3 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 8)$. What is its y-coordinate when $x = 1$?

a. 8

b. 14

c. 12

d. 16

Procedural Fluency & Algebraic Manipulation

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 In $\triangle ABC$ studying proving triangles congruent, $m\angle A = 31^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 97°

b. 107°

c. 89°

d. 149°

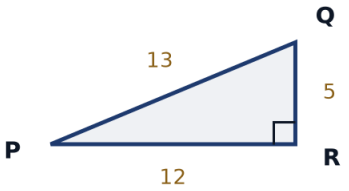
5 A geometric figure scaled by factor $k = 3$ has an original side length of 5 cm. What is the corresponding image length?

a. 12 cm

b. 15 cm

c. 20 cm

d. 45 cm



6 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 8)$. What is its y-coordinate when $x = 2$?

a. 12

b. 18

c. 16

d. 20

Graphical & Geometric Interpretation

7 In $\triangle ABC$ studying proving triangles congruent, $m\angle A = 31^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 90°

b. 100°

c. 82°

d. 149°

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

8 A geometric figure scaled by factor $k = 4$ has an original side length of 5 cm. What is the corresponding image length?

a. 16 cm

b. 20 cm

c. 25 cm

d. 80 cm

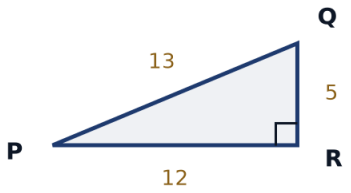
9 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 8)$. What is its y-coordinate when $x = 3$?

a. 16

b. 22

c. 20

d. 24



Exam-Style Problem Solving & Modeling

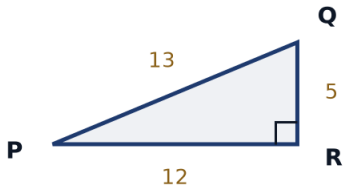
10 In $\triangle ABC$ studying proving triangles congruent, $m\angle A = 31^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 83°

b. 93°

c. 75°

d. 149°



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

11 A geometric figure scaled by factor $k = 5$ has an original side length of 5 cm. What is the corresponding image length?

a. 20 cm

b. 25 cm

c. 30 cm

d. 125 cm

12 On the coordinate plane, a line with slope $m = 4$ passes through $(0, 8)$. What is its y-coordinate when $x = 4$?

a. 20

b. 26

c. 24

d. 28

© ClipSAT: By Mr. Mohamed Abdallah +966597688647